Newton's First Law

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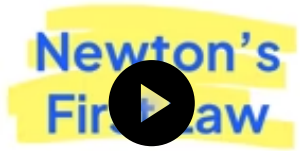

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Newton's first law

- Newton's **1st law of motion** states:

Objects will remain at rest, or move with a constant velocity unless acted on by a resultant force

- This means if that:
  - A **stationary** object will **remain** stationary
  - An object **moving** with a **constant** velocity will continue to move at that constant velocity
  - Unless the object is acted upon by a **resultant force**
- Conversely, that means that if an object is at rest, or has a constant velocity, **no** resultant force **can** be acting on it
- Remember that a **constant velocity** means:
  - No change in **speed**
  - No change in **direction**

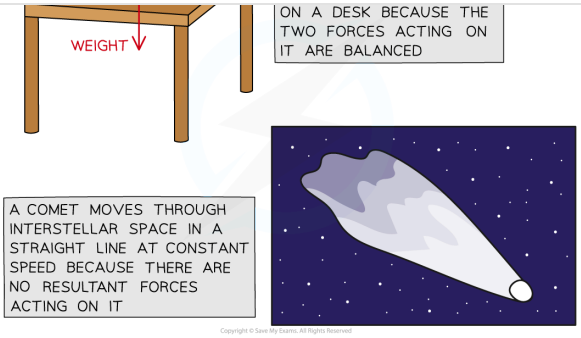
Objects with Zero Resultant Force

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Constant velocity can only be achieved when the forces on an object are balanced – in other words, when the resultant force is zero

Examples of Newton's first law

- A mug on a table will remain stationary on the table unless acted upon by a resultant force
  - For example, if someone picks up the mug, or knocks into the table
- A piece of space debris will continue moving in a straight line at a constant speed unless acted upon by a resultant force
  - For example, if it enters the gravitational field of a planet or collides with an asteroid
- A car travelling in a straight line at a constant speed will continue to travel in a straight line at a constant speed unless acted upon my a resultant force
  - For example, if the driver brakes or accelerates

Worked Example

A student did some online research and found out that the Moon orbits the Earth at a constant speed of around 2000 mph.

The student says that this is not an example of Newton's first law of motion. Is the student correct? Explain your answer.

Answer:

Step 1: Recall Newton's first law of motion

- Newton's first law of motion states that **objects will remain at rest, or move with a constant velocity, unless acted on by a resultant force**

Step 2: Determine if the object in the question is at rest, or if it is moving with a constant velocity

- The Moon, in this case, is **not** at rest
- It is moving at a **constant speed**
- But it is **not** moving in a **constant direction** – it continually orbits the Earth

Step 3: State and explain whether the student is correct

- The student is **correct**
- The Moon moves with a **constant speed**, but always **changes direction**
- So it is **not** moving with a **constant velocity**, and is **not** an example of Newton's first law of motion

Worked Example

A car moves at a constant velocity. The driving force from the engine is 3 kN.

Determine the frictional force acting on the car.

Answer:

Step 1: Recall Newton's first law of motion

- Newton's first law of motion states that **objects will remain at rest, or move with a constant velocity unless acted on by a resultant force**

Step 2: Relate Newton's first law to the scenario

- Since the car is moving at a constant velocity, there is no resultant force
- This means the driving and frictional forces are balanced

Step 3: State the value of the frictional force

- Frictional force, F = driving force = **3 kN**

Examiner Tips and Tricks

Students often struggle to understand how an object can move when there is no resultant force acting on it. You may even have this misconception without realising it.

It is important to fully understand Newton's first law, because it underpins a lot of physics content and can cause you to make mistakes in seemingly unrelated questions.

Usually, students are comfortable with the idea that an object at rest on a table will not move unless something or someone physically moves it. The issue comes from the idea of an object in motion.

A rock drifting through the vacuum of deep space away from the gravitational pull of any large bodies, will continue to drift

On Earth, there are always forces acting on objects, so in our everyday experience we need to think in terms of resultant forces and balanced forces.

A car can maintain a constant velocity when the force from the engine is balanced with the frictional forces opposing its motion (friction between the tyres and road, and air resistance). If nothing changes with these forces, it will continue travelling at a constant velocity. If you do want to change the motion of the car, to speed up or slow down, you need to provide a resultant force. By increasing the force from the engine, you can cause the car to speed up. By either decreasing the force from the engine, or increasing the frictional force (using the brakes) the car will slow down.

When the forces acting on an object are not balanced (a resultant force is exerted) this will cause a change in the object's motion.

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